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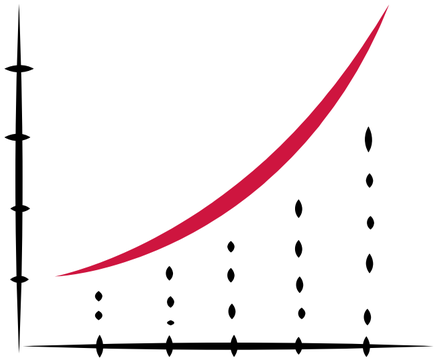
$f\theta(X) \approx y$

MACHINE LEARNING FUNDAMENTALS



From raw data to a model “you can trust”

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29/09/2026

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ROADMAP

Did the Machine Really Learn?

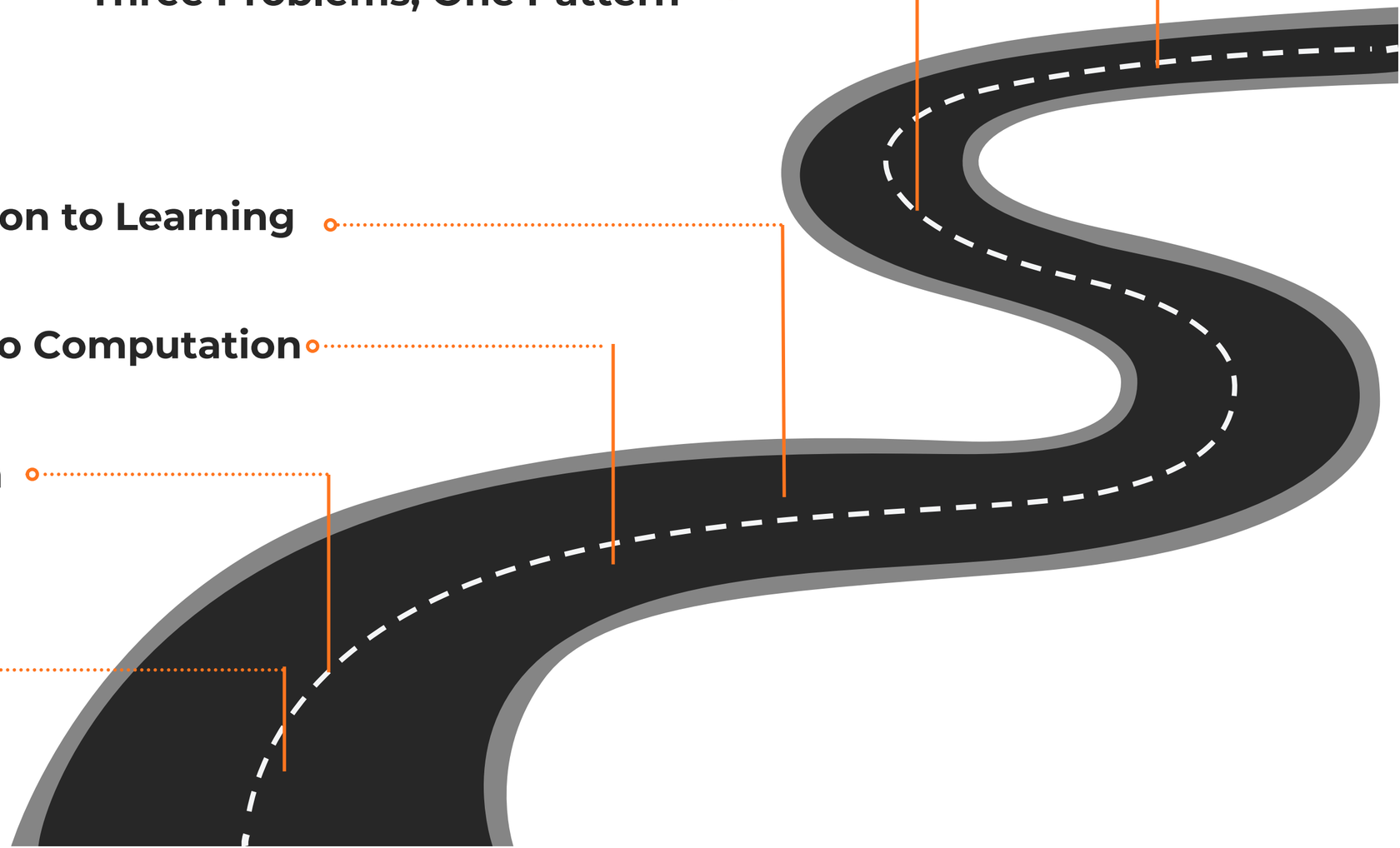
Three Problems, One Pattern

From Computation to Learning

From Data to Computation

From Reality to data

The Landscape



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01

THE AI LANDSCAPE

Where AI stands today, and why ML still matters underneath it all

A BRIEF HISTORY OF AI

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1956

“AI” is coined

The term is born

1980s

Expert systems

Rule-based logic for medical & business decisions

2012

Deep learning returns

AlexNet shocks the vision world — neural nets are back

2017

Transformers

“Attention Is All You Need” changes everything

2022

ChatGPT

LLMs go mainstream — 100M users in 60 days

2024+

Agents

Models that plan, call tools, and act autonomously

WHERE DOES ML ACTUALLY STAND?

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ARTIFICIAL INTELLIGENCE , systems that act intelligently

Symbolic AI

Hand-coded rules, search & planning

- Expert systems
- Game-tree search
- Route planning

Older paradigm — logic supplied by a human, not learned

Machine Learning

Learns patterns from data, instead of being told the rules

Deep Learning

ML with layered neural networks

LLMs & Agents

Today's application layer: language models + tool use + planning loops, built ON TOP of deep learning

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WHY ML STILL MATTERS ?

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02

FROM REALITY TO DATA

Data Holds **Patterns**

A MYSTERY, NOT A DEFINITION

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Study h	Attend.	Sleep	Score
2	55%	5h	42
7	91%	7h	84
5	80%	6h	71
1	65%	8h	48
8	95%	6h	89

Hours studied · Attendance · Sleep · Exam score — five rows, four columns.

"Students who study more score higher."

Is that written anywhere in this table?

- Not explicitly. The table holds data.
- **You produced the knowledge — a human, reading rows.**

DATA ≠ INFORMATION ≠ KNOWLEDGE

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DATA

`7 · 91 · 6 · 84`

Raw observations — meaningless alone

INFORMATION

`study_hours=7, attendance=91%, score=84`

Data plus structure

KNOWLEDGE

`More study & attendance → higher scores`

A usable pattern in this set

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03

FROM DATA TO COMPUTATION

Turning messy reality into something a machine can compute on.

EDA

```
df.head()
df.describe()
df.isna().sum()
df["exam_score"].hist()
df.corr(numeric_only=True)
```

- What variables exist, and what do they actually mean?
- What values are typical — what counts as unusual?
- What's missing, and is it missing at random?
- Could the data itself be quietly misleading you?

WHY EDA IS WHERE PROJECTS FAIL

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- **Missing values** that silently bias the model
- **Wrong types** “7” the string vs. 7 the number
- **Outliers** that are typos — or the most important rows in the set
- **Duplicates & mismatches** inconsistent units, duplicate rows, mismatched dates
- **Leakage** information from the future sitting inside training data

WHY PYTHON?

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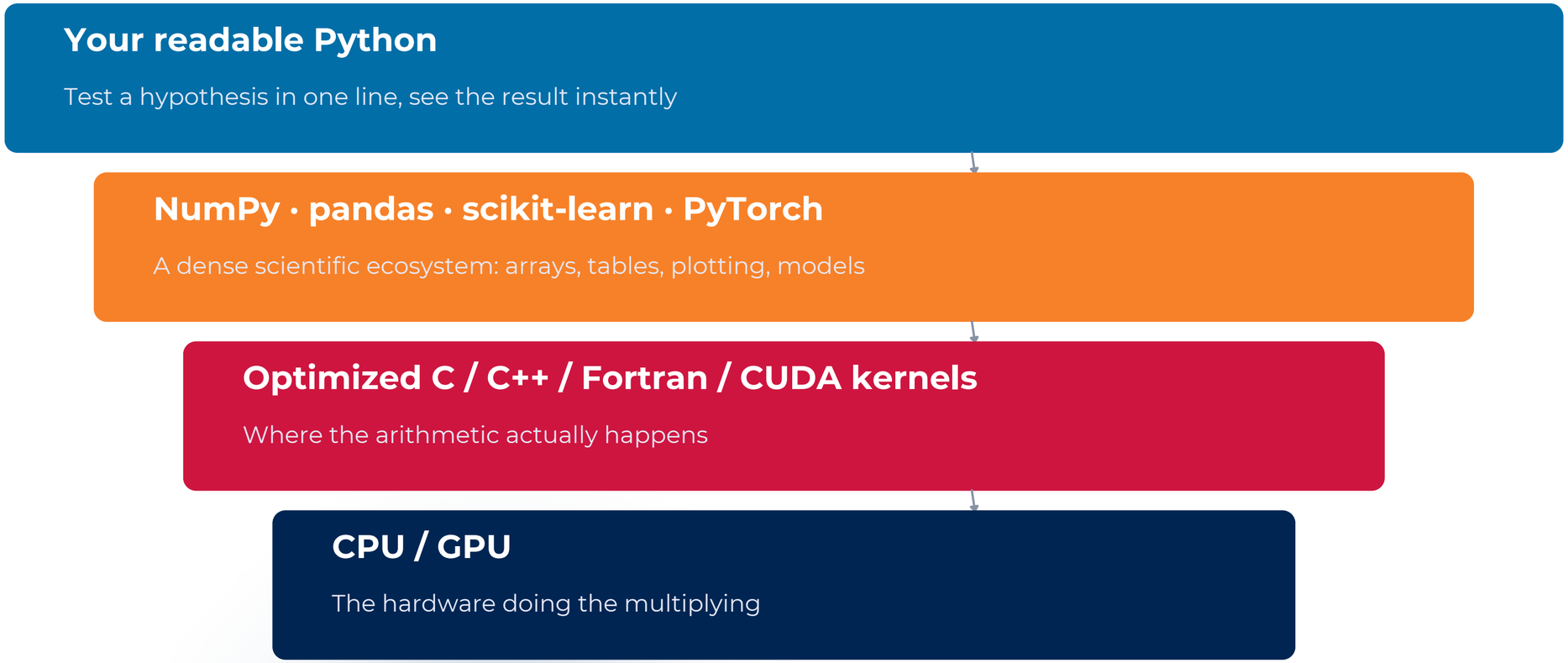

VECTORS & MATRICES

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Real datasets have thousands to millions of rows. You could loop through numbers one at a time — but that fights both the math and the machine.

THE NAIVE WAY — numbers, one at a time

```
pred = 0
for i in range(len(w)):
    pred += x[i] * w[i]
pred += b
```

THE VECTOR WAY — one operation, whole row

```
pred = x @ w + b
```

the dot product IS the weighted sum — same math, one line

Mathematical fit

Means, distances, and predictions ARE linear algebra. $\hat{y} = Xw$ is literally one matrix multiplication — not a nested loop pretending to be one.

Computational fit

A vector is stored as one contiguous block. Hardware processes contiguous blocks far faster than scattered Python numbers — next slide shows exactly why.

ONE OPERATION

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Task: add 1 to one million values.

SCALAR / LOOP LOGIC

```
for i in range(len(x)):
    x[i] = x[i] + 1
```

Python asks the interpreter to manage each element, one iteration at a time.

1,000,000 Python-level iterations

VECTOR / ARRAY LOGIC

```
x = x + 1
```

One high-level instruction is handed to an optimized compiled array kernel.

1 Python call → bulk numerical work

The CPU still performs every one of those additions
vectorization does not eliminate the loop. It moves WHERE the loop runs: out of slow Python-level interpretation, into compiled numerical code that already knows how to use cache and SIMD.

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04

FROM COMPUTATION TO LEARNING

Where fitting a function becomes “learning.”

SUPERVISED VS UNSUPERVISED

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Have y → SUPERVISED

Number to predict → Regression

Example: apartment price, given size & location

Category to predict → Classification

Example: pass / fail, given study hours

Every row already has the correct answer attached — the model learns to reproduce it on new rows.

No y → UNSUPERVISED

Find structure in X alone → Clustering

Example: group members by shared interests, no labels given

There is no correct answer to imitate — the model has to discover the structure on its own.

THE ML PROBLEM

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$$f\theta(X) \approx y$$

- You have X and y. You don't know the true function between them.
- θ — the parameters we're free to adjust.
- Learning = find θ so that f gets close to reality.

WHAT DOES LEARNING MEAN

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We don't want a function that only remembers the training table. We want one that works on new observations.

WE OBSERVE A SAMPLE

$$D = \{(x_i, y_i)\}^n_{i=1}$$

A finite dataset — one draw from that distribution, with n examples.

LEARN PARAMETERS θ

$$f\theta(X) \approx y$$

Training asks: find θ that makes the loss small on the examples we observed.

LEARNING ≠ ALGORITHM ≠ MODEL

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ALGORITHM

adjusts w, b
to reduce error

MODEL

$\hat{y} = 8.1x + 21$
the learned object

trained model → prediction on unseen X

THE LOSS

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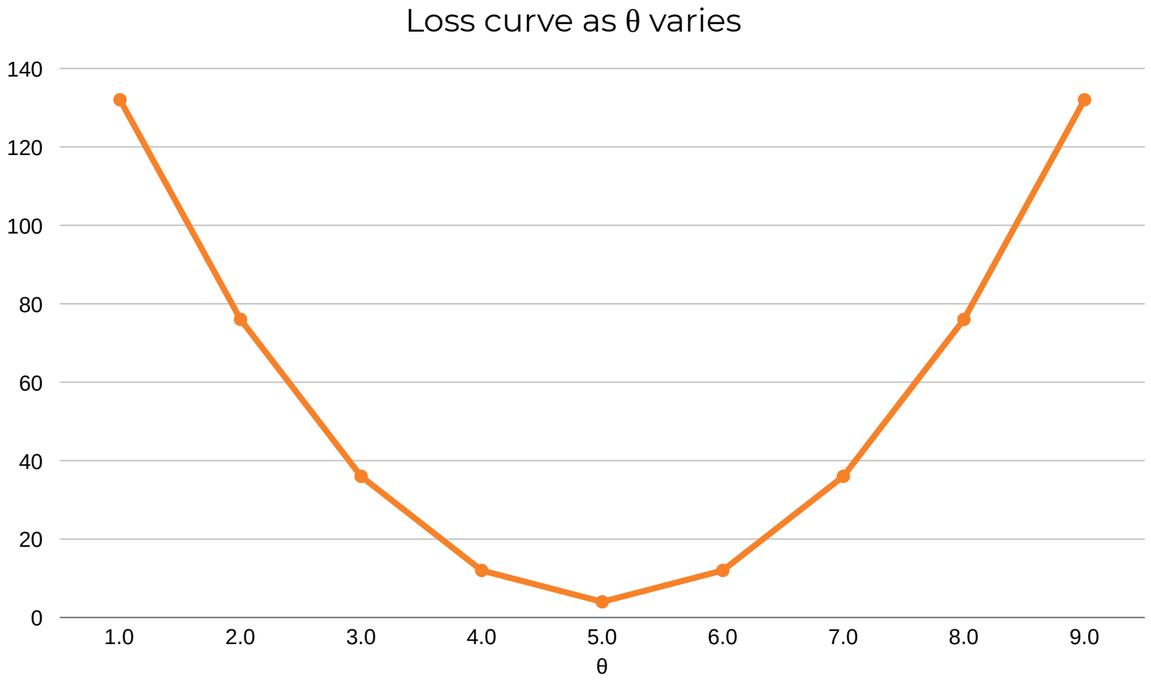
Real y	Predicted ŷ	Error y−ŷ	Squared
120	110	10	100
160	172	-12	144
210	202	8	64

$$L(w,b) = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$$

GRADIENT DESCENT



1 **Predict**

2 **Loss**

3 **∂L/∂θ**

4 **Update θ**

current θ, and −∇L points downhill toward the minimum

THE LEARNING LOOP

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A model learns by repeatedly turning parameters into predictions, measuring error, then updating the parameters.

1 **Parameters θ**

2 **Predict $\hat{y} = f_\theta(X)$**

3 **Measure loss**

4 **Update θ**

repeat until improvement stops / budget ends

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05

MODELS

OVERVIEW

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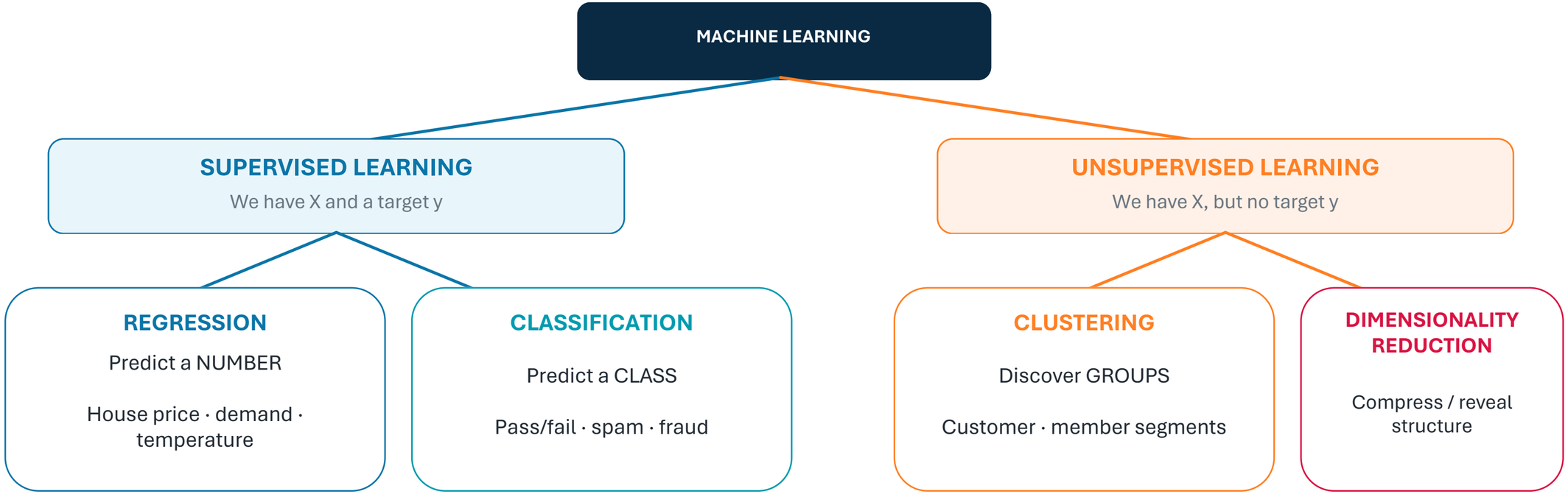

EXAMPLES OF ALGORITHMS

- Linear Regression
- Logistic Regression
- Decision Trees / RF
- K-Means
- PCA

EXAMPLES

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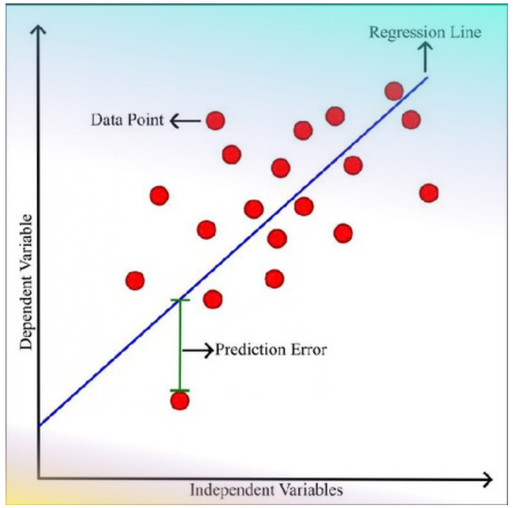
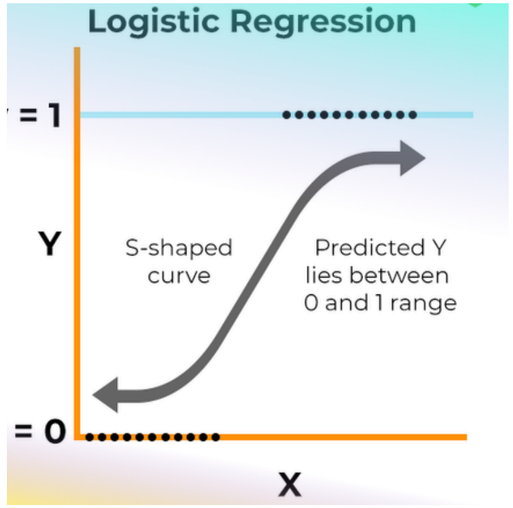
Predict a quantity

Linear Regression



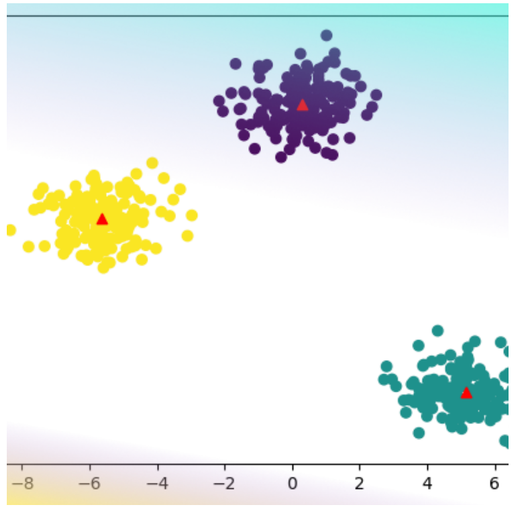
Predict a category

Logistic Regression



No labels at all

K-Means Clustering



TRAIN VS TEST:

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TRAIN SET

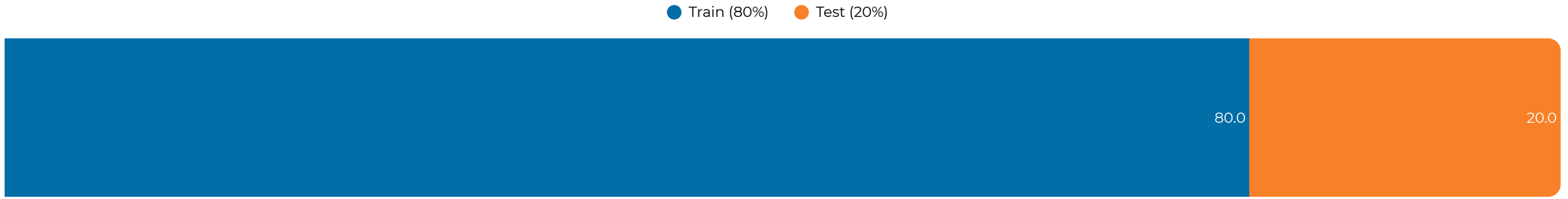
Solved exercises — the model learns here

If you memorize the correction, you can look perfect on training.

TEST SET

New exam question — left completely untouched

The model is evaluated only on this unseen set.



UNDERFITTING VS OVERFITTING

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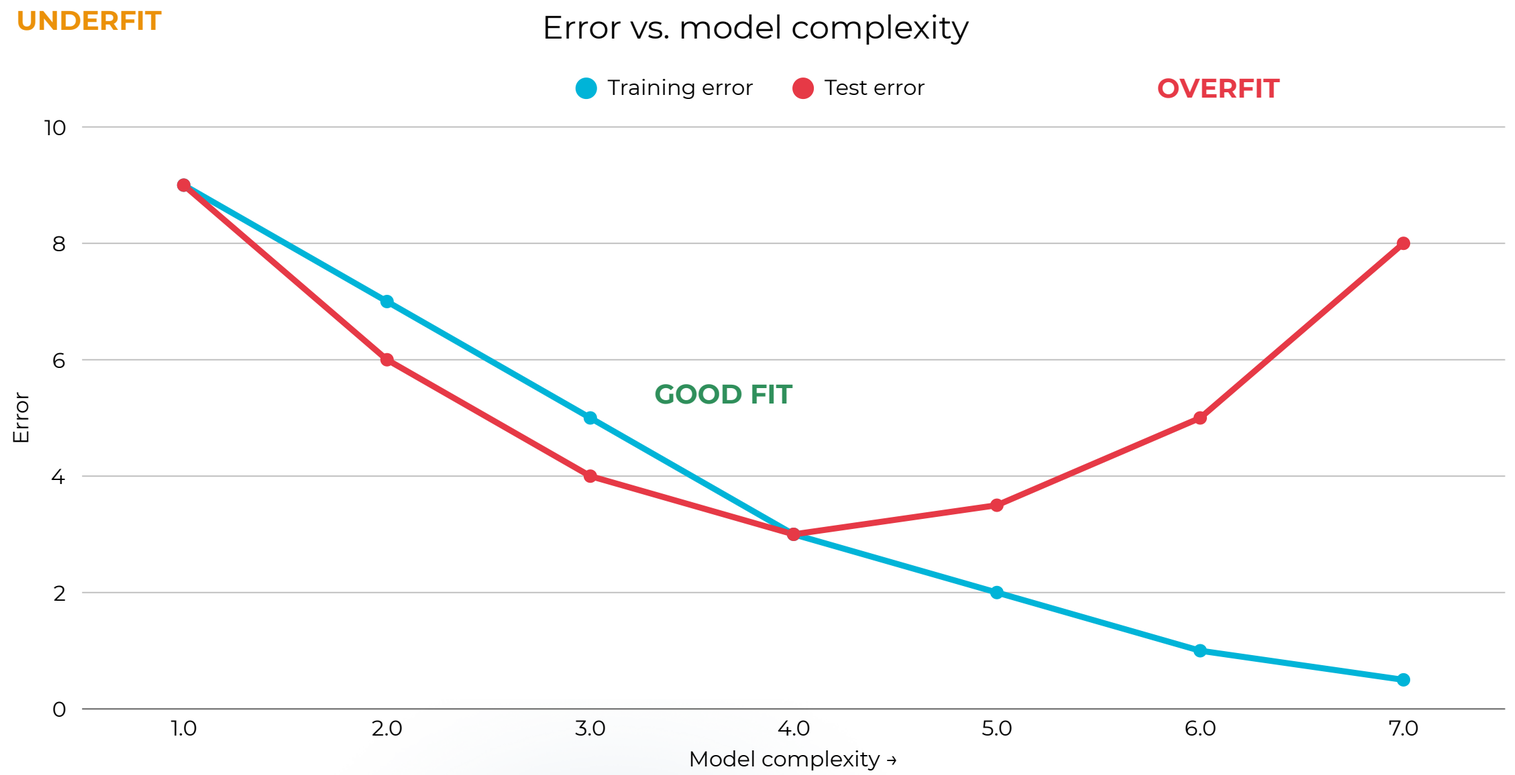

Underfitting

Model too simple. Train error high, test error high.

Good fit

Captures the real signal. Both errors are low.

Overfitting

Memorizes training noise. Train near zero, test high.

Goal ≠ minimize training error at any cost. Goal = perform well on unseen data.

THE EVALUATION TRAP

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100 patients: 95 normal, 5 abnormal.

	Pred +	Pred −
Actual +	**TP = 3**	**FN = 2**
Actual −	**FP = 0**	**TN = 95**

Accuracy = 98% here — but 2 of the 5 real positives were missed entirely.

Is that an excellent model? Obviously not.

A metric only matters relative to which errors actually cost you.

THE WORKFLOW

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Real-world question → **Data collection** → **EDA + statistics** → **Data preparation** → **Train / test split** → **Choose & train model** → **Predict & evaluate** → **Generalize? improve** ↺

```
X = df[["study_hours","attendance"]]
y = df["passed"]

X_tr, X_te, y_tr, y_te = train_test_split(
    X, y, test_size=.2)

model = LogisticRegression()
model.fit(X_tr, y_tr)
pred = model.predict(X_te)
```

NumPy
vectors, matrices, numerical computation

pandas
tables, cleaning, EDA

scikit-learn
preprocessing, training, evaluation

COLAB TIME

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THANK YOUUU

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